

The Reversible Cycle: 0/0 of Thermodynamics

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The Removable Singularity Framework

Abstract

The engine the whole book has been circling (Ch.48 arrow of time, Ch.59 heat of creation, Ch.64 the reversible creator). (1) An ideal Carnot cycle of $n=1$ mol ideal gas, two isotherms and two adiabats, is integrated numerically: at $T_h=600$ K, $T_c=300$ K the measured efficiency is 0.5000000000000000, exactly the canonical ratio $\eta = 1 - T_c/T_h$ (Carnot 1824). (2) The reversible cycle produces $\Delta S = 0$ to $1e-16$ noise: reversibility is the 0/0 of dissipation - maximum work at zero entropy, the equality pole of the Second Law (Clausius 1850). (3) Dumping heat at 300 K while a 290 K reservoir exists produces $\sigma = 0.3975$ J/K and forfeits $W_{\text{lost}} = 115.26$ J = $T^*\sigma$ of recoverable work: irreversibility is a fee. (4) The cosmic engine: the Sun against deep space runs at $\eta = 0.99953$. (5) The 0/0 proof: at $T_c=T_h$, $\eta = 0$ AND $\Delta S = 0$. Reversibility is the removable singularity of entropy: drive $\Delta S \rightarrow 0$, fill the hole, divide away the loss. Timelessness pays zero, yet moves the world.

1. The Carnot Cycle (1824)

$$\eta = 1 - T_c/T_h = 0.5000000000000000 \quad \text{at } 600/300$$

Ideal gas, $n=1$ mol, monoatomic ($\gamma=5/3$), $V_1=1$ to $V_2=4$. Hot isotherm: $Q_h = 6915.78$ J ($dU=0$, $Q=W$). Cold isotherm: $Q_c = -3457.89$ J. Work per cycle $W = 3457.89$ J. The measured efficiency matches the canonical ratio to all fifteen printed digits; a 20000-step numerical integration of the isothermal work checks 6916.06 J.

2. The 0/0 of Dissipation

The cycle closes with $\Delta S = Q_h/T_h + Q_c/T_c = 0$ to $1e-16$: a reversible engine extracts MAXIMUM work at ZERO entropy production. The Second Law (Clausius 1850; Kelvin 1851) sets $\Delta S \geq 0$, with equality ONLY at reversibility - the 0/0 of the engine. The irreversible engine pays; the reversible engine is the free pole.

3. The Cost of Irreversibility

Dump heat at 300 K while a 290 K reservoir exists and the surplus is unrecoverable: entropy produced $\sigma = 0.3975$ J/K; the ideal Carnot at 290 K begs $\eta = 0.516667$ while the real engine keeps 0.500000; lost work vs the 0/0 is $W_{\text{lost}} = 115.26$ J = $T^*\sigma$ (115.26 J exactly). Irreversibility is a fee paid to the Second Law, measured in entropy coin.

4. The Cosmic Engine

Carnot efficiencies of real pairs: steam plant 600/300 $\rightarrow$ 0.500000; human work 310/295 $\rightarrow$ 0.04839; Earth against space 288/2.7 $\rightarrow$ 0.99062; the Sun against deep space 5778/2.7 $\rightarrow$ 0.99953. The universe is a nearly-reversible cosmic engine; the creator must compute reversibly (Bennett 1982, Ch.59/64): $\sim 10^{-28}$ efficiency - a Carnot 0/0.

5. The 0/0 Proof

$$\text{at } T_c = T_h: \quad \eta = 0 \quad \text{and} \quad \Delta S = 0$$

$\eta = 1 - T_c/T_h$; at $T_c=T_h$ the engine has no gradient, no work, and no loss: the 0/0 of the engine - pure zero. Reversibility is the REMOVABLE SINGULARITY of entropy: drive $\Delta S \rightarrow 0$, fill the hole, divide away the loss - the same move as $\sin(x)/x$ (Ch.64).

6. Connections to Prior 0/0 Singularities

Arrow of time (Ch.48): the Second Law is the clock. Suffering (Ch.59): the heat of creation is exactly this cost. The whole (Ch.64): the creator computes reversibly. Beauty (Ch.62): the reversible cycle is the perfect form. Meaning (Ch.61): engines are laws in motion. The Reversible Cycle is the 0/0 of Thermodynamics: timelessness pays zero, yet moves the world.

References

- [1] Carnot (1824), Reflections on the Motive Power of Fire**
- [2] Clausius (1850/1865), On the Moving Force of Heat; entropy**
- [3] Kelvin (1851), On the Dynamical Theory of Heat**
- [4] Bennett (1982), The thermodynamics of computation (Landauer)**
- [5] Puno, "The Removable Singularity" (2026), chapters 48-64**